In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST - IV (PAPER-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**A**	**B**
2.	**A**	**C**	**B**
3.	**C**	**A**	**B**
4.	**B**	**A**	**C**
5.	**A**	**B**	**D**
6.	**ABCD**	**BC**	**AB**
7.	**BC**	**ABD**	**ABD**
8.	**AD**	**ABCD**	**BD**
9.	**ACD**	**ABC**	**BC**
10.	**ABC**	**ABCD**	**AC**
11.	**AC**	**ABC**	**AC**
12.	**BC**	**ABC**	**BD**
13.	**ACD**	**CD**	**AC**
1.	**1**	**6**	**2**
2.	**3**	**5**	**3**
3.	**7**	**4**	**3**
4.	**9**	**8**	**7**
5.	**5**	**6**	**4**

Physics

PART – I

SECTION – A

1. $O=(25)^2-2ax.....(1)$

$\frac{-3}{10}t=\frac{dv}{dt}$

$\int_{15}^{v} dv=\frac{3}{10}\int_{0}^{t} t\ dt$

$v=15-\frac{3}{20}t^2.....(2)\ when\ v=0\Rightarrow t=10\sec$

$\int_{0}^{225-x} dx=15\int_{0}^{10} dt\ \frac{-3}{20}\int_{0}^{10} t^2\ dt$

$225-x=15(10)-\frac{3}{20}\frac{(1000)}{3}$

$225-x=150-50.....(3)$

Solving (1) and (3)

$a=2.5\,m/s^2$

2. $M_1 g\sin\theta>\frac{M_1 g\cos\theta}{5}$

M_1 will slip if $\tan\theta>\frac{1}{5}$

$M_2 g\sin\theta'+\frac{M_1 g\sin\theta'}{5}>\frac{1}{2}M_2 g\cos\theta'$

M_2 will slip if $\tan\theta'>\frac{1}{2+\frac{2}{5}\frac{M_1}{M_2}}$

So M_1 will slip first

3. Using momentum conservation

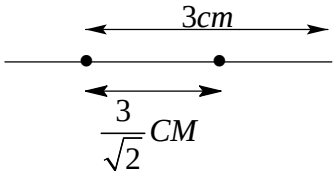
$m\sqrt{\frac{GM}{2R}}=\frac{m}{2}V'$

$V'=\sqrt{\frac{2GM}{R}}$

$\Delta U=\left(\frac{1}{2}\frac{m}{2}\frac{2GM}{R}\right)-\left(\frac{1}{2}m\frac{GM}{2R}\right)$

$\Delta U=\frac{mgR}{4}$

4.



Time period $= 4\sec$

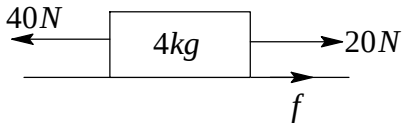
Distance travelled in $8\sec = 24cm$

Distance travelled in $8.5\sec = 24 + 3 - \frac{3}{\sqrt{2}} = \left(27 - \frac{3}{\sqrt{2}}\right)cm$

5. $k = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{a}$

$q = \sqrt{4\pi\varepsilon_0 ak}$

6.



$f = 20N\left(< 24N\right)$

So block will not move

Contact force $= \sqrt{(20)^2 + (40)^2} = 20\sqrt{5}\,N$

7. Option (a) and (d) violate law of conservation of momentum.

8. Buoyancy force when lift accelerates upwards $= V_s(g + a)\rho$

$V\sigma g = V_s\delta g.....(1)$

$V_s'\rho(g + a) - V\sigma g = V\sigma a$

$V_s'\rho g + V_s'\rho a - V\sigma g = V\sigma a$

$V_s'\rho g + V_s'\rho a - V_s\rho g = V_s\rho a$ $\left(\text{Using equation}(1)\right)$

$V_s'(g + a) = V_s(a + g)$

$V_s' = V_s$

10. Flux depends upon charges inside Gaussian surface and field on all charges.

11. For same sense field $B = \frac{N_1\mu_0 I}{2R_1} + \frac{N_2\mu_0 I}{2R_2} = 8\pi \times 10^{-4}T$

For opposite sense field $B' = \frac{N_1\mu_0 I}{2R_1} = \frac{N_2\mu_0 I}{2R_2} = 0$

12. $V_1^{'} + V_2^{'} = \dfrac{Ldi}{dt}$

When current is maximum $\dfrac{di}{dt} = 0 \Rightarrow V_1^{'} = -V_2^{'}$

And $\varepsilon = \dfrac{Ldi}{dt} = 0$

SECTION – C

3. $\dfrac{qb}{2\pi R} ER = \dfrac{1}{2} mR^2\omega^2$

$\omega = \sqrt{\dfrac{7qEb}{22mR^2}}$

So $C = 7$

4. $y = \dfrac{1/A}{x/\ell} .....(1)$

$y = \dfrac{F/3A}{x/3\ell} .....(2) \Rightarrow F = 9N$

5. $T_1 = 2\pi\sqrt{\dfrac{1.44}{g}} = 1.2\ T_2\ \ where T_2 = \dfrac{2\pi}{\sqrt{g}}$

$\dfrac{T_1}{T_2} = \dfrac{6}{5}$

So after 5 oscillation of first they will swing in phase

Chemistry

PART – II

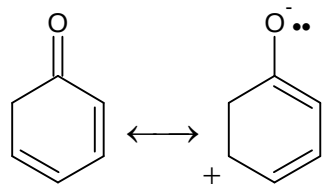
SECTION – A

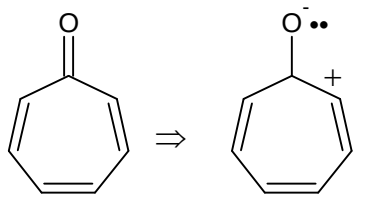
1. In option (a) all methyl groups are equatorial

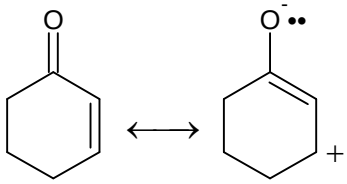
2. (I) is trans and (II) is cis form

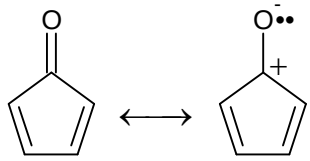
3. $\underbrace{2KI}_{\text{Four ions}} + HgI_2 \longrightarrow \underbrace{K_2[HgI_4]}_{\text{three ions}}$

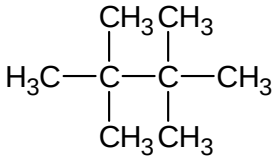
4. Due to back bonding (a) is a weaker base.

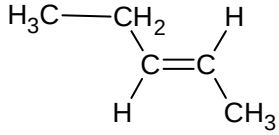
6. (a) 

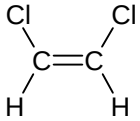
(b) 
aromatic

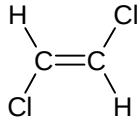
(c) 

(d) 
anti aromatic

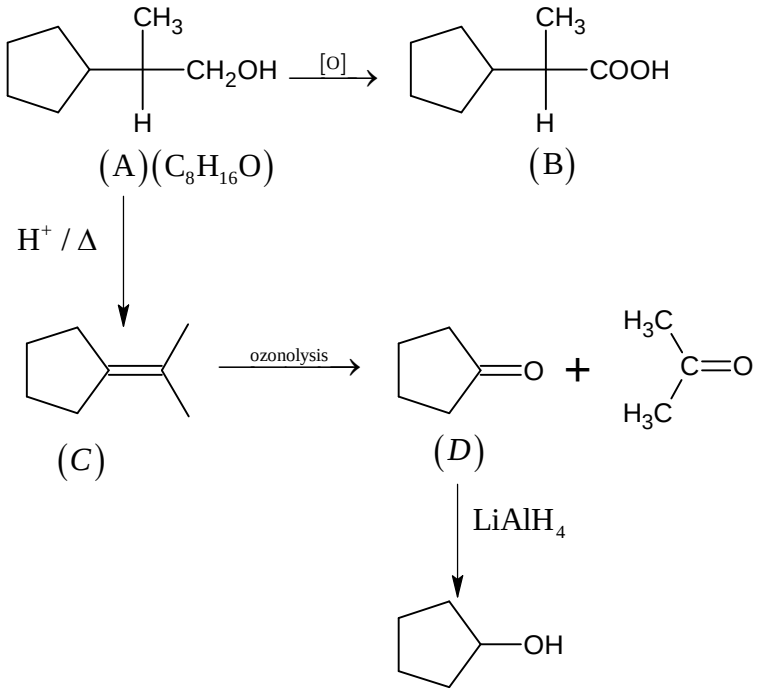
9. (a) 

(b) 

(c) 

(d) 

10.

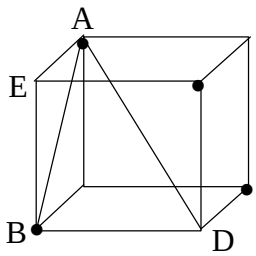


11. An amino acid at its isoelectric point, will not migrate.

12. A lone pair of N-atom participating in resonance will be less basic

SECTION – C

1.



Face diagonal of cube $(AB)=2R$

Edge length of cube $(BE)=\frac{2R}{\sqrt{2}}$

Body diagonal of cube $(AD)=\sqrt{3}\times\frac{2R}{\sqrt{2}}$

$=\sqrt{6}\times R$

2. $Mg^{2+}+2e^{-}\longrightarrow Mg\left(E_{Mg}=\frac{24}{2}=12g\right)$

Charge required for the deposition of $12g\ Mg=1F$

Charge required for the deposition of $1g\ Mg = \frac{1}{12}F$

$\therefore$ cost for $\frac{F}{12}$charge=Rs.5

$Al^{3+} + 3e^- \longrightarrow Al\left(E_{Al} = \frac{27}{3} = 9g\right)$

Charge required for $9\,g = 1F$

Charge required for $30\,g = \frac{30}{9}F = \frac{10F}{3}$

Cost for $\frac{10F}{3}$charge$= 5 \times 12 \times \frac{10F}{3}$

$= Rs.200$

3. $pKa\ of\ HX = 14 - 10 = 4$

$pH = pK_a + \log\frac{1}{1} = 4$

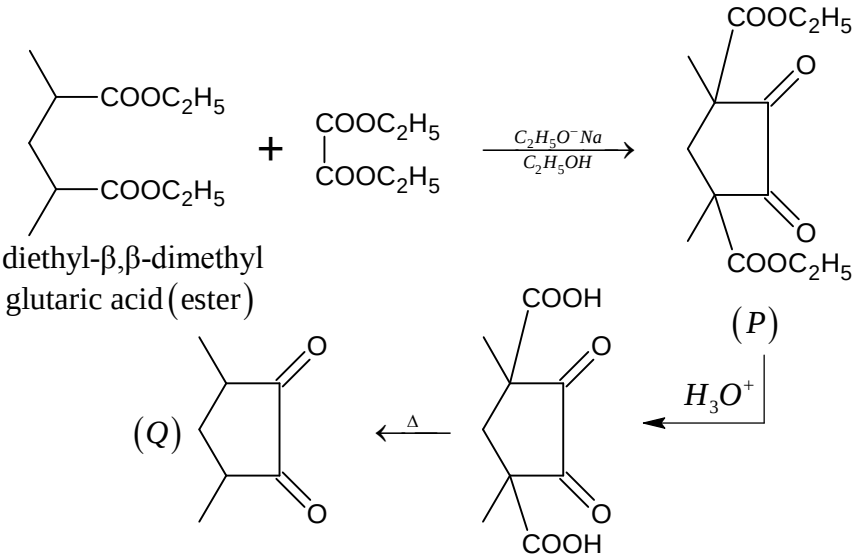
4. $(CO)_{x/2}\,Co - Co(CO)_{x/2}$

$EAN\,(=36) = 27 + 1 + 2 \times \frac{x}{2}$

$\Rightarrow 28 + x = 36$

$x = 8$

5.



Mathematics

PART – III

SECTION – A

1. $\dfrac{f''(x)}{f'(x)} - \dfrac{g''(x)}{g'(x)} = \dfrac{\dfrac{d}{dx}\left(f'(x)/g'(x)\right)}{\dfrac{f'(x).g'(x)}{(g'(x))^2}} = \dfrac{g'}{f'}.\dfrac{d}{dx}\left(\dfrac{f'}{g'}\right) = \dfrac{2f'(x)}{f(x)}\left[\because f(x).g(x) = 1\right].$

2. Let $f^{-1}(x) = t \Rightarrow f(t) = x$

$\displaystyle\int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx = \int_a^b 2f(t).(b-t).f'(t)dt$ = b $[(f(t))^2]_a^b$ - b $\displaystyle\int_a^b$ t .2f(t) .f '(t) dt

$= \displaystyle\int_a^b \left[(f(x))^2 - (f(a))^2\right]dx$

3. $y = \log_{1/3}\left(x - \dfrac{1}{3}\right) + \log_3\sqrt{9x^2 - 6x + 1} = \log_{1/3}\left(x - \dfrac{1}{3}\right) + \log_3|3x - 1| = 1.$

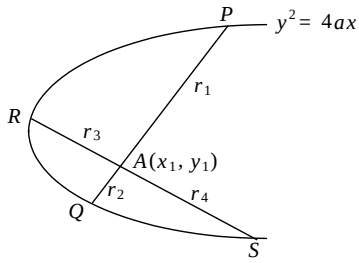
Since $x > \dfrac{1}{3}$, maximum {AB} = 3.

4. $a_n = \dfrac{\dfrac{3n}{2}\left[a + (3n-1)d\right]}{\dfrac{2n}{2}\left[a + (2n-1)d\right] - \dfrac{n}{2}\left[a + (n-1)d\right]} = 3; \quad \therefore \underset{n\to\infty}{\text{Lt}} \displaystyle\sum_{r=1}^{2n} \frac{1}{n}.3 = 6.$

5. $e^{2x} + 2e^x < a(e^x - 1) \Rightarrow (e^x - 1)^2 + 4(e^x - 1) + 3 < a(e^x - 1)$

$\Rightarrow (e^x - 1) + \dfrac{3}{e^x - 1} + 4 < a \Rightarrow a > 4 + 2\sqrt{3}.$

7. Let two perpendicular chords through $A(x_1, y_1)$ be PQ and RS.



Equation of PQ is $\dfrac{x - x_1}{\cos\theta} = \dfrac{y - y_1}{\sin\theta} = r$ where $\tan\theta$ = slope of PQ. Any point on this line may be taken as $(x_1 + r\cos\theta, y_1 + r\sin\theta)$. As the point lies on $y^2 = 4ax$,

$\therefore \quad (y_1 + r\sin\theta)^2 = 4a(x_1 + r\cos\theta).$

$\Rightarrow \quad r^2\sin^2\theta + 2y_1\sin\theta.r - 4a\cos\theta r + y_1^2 - 4ax_1 = 0$

$\therefore \quad r_1 r_2 = \dfrac{-(y_1^2 - 4ax_1)}{\sin^2\theta}$

Similarly $r_3 r_4 = \dfrac{-(y_1^2 - 4ax_1)}{\sin^2\alpha}$ where $\tan\alpha$ = slope of RS

Since PQ is perpendicular to RS,

$\therefore \quad \alpha = 90 + \theta$ or $\theta - 90$

In either case; $\sin^2\alpha = \cos^2\theta$

Now $\frac{1}{r_1 r_2}+\frac{1}{r_3 r_4}=\frac{\sin^2\theta+\cos^2\theta}{-\left(y_1^2-4ax_1\right)}=\frac{1}{-\left(y_1^2-4ax_1\right)}$ which is constant

8. Curve is $(y-2)^2 = 2x^3 - 4$ (1)

$$\frac{dy}{dx}=\frac{3x^2}{y-2}$$

$\therefore$ Equation of tangent at (x_1, y_1)

$$y-y_1=\frac{3x_1^2}{y_1-2}\left(x-x_1\right)$$

As it passes through (1, 2)

$$\Rightarrow \quad 2-y_1=\frac{3x_1^2}{y_1-2}\left(1-x_1\right)$$

or $\left(y_1-2\right)^2=3x_1^2\left(x_1-1\right)$

or $2x_1^3-4=3x_1^3-3x_1^2$ [As (x_1, y_1) lies on the curve]

or $x_1^3-3x_1^2+4=0$

or $(x_1 + 1)(x_1 - 2)^2 = 0$

$\therefore$ $x_1 = -1$ or 2

But $x_1 \neq = -1$ (it is clear from equation (1))

$\therefore$ $x_1 = 2$

$\therefore$ $y_1 = 2 \pm 2\sqrt{3}$

Points of contact are $(2, 2 + 2\sqrt{3})$ and $(2, 2 - 2\sqrt{3})$

$\therefore$ sum of x–coordinates of points of contact is 4

$\therefore$ sum of y–coordinates of points of contact is 4.

9. If $\left|\int_a^b f(x)dx\right|=\int_a^b |f(x)|dx$

$\Rightarrow$ $f(x)$ is either positive in (a, b)

or $f(x)$ is negative in (a, b) and also $b > a$

Also $g'(x)=f(x)$

When $f(x) > 0 \Rightarrow g'(x) > 0 \Rightarrow g(x)$ is increasing function

When $f(x) < 0 \Rightarrow g'(x) < 0 \Rightarrow g(x)$ is decreasing function

also $g(0) = 0$

$$\int_a^b f(x)g(x)dx=\int_a^b g(x)g'(x)dx$$

$$= \frac{\left(g(b)\right)^2-\left(g(a)\right)^2}{2}$$ which will be positive for $0 < a < b$.

10. $C_1 \to C_1 - C_2$

$$\Rightarrow \Delta = \begin{vmatrix} 4 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ 4 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ 4 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix} = 0 \qquad (\because C_1 \text{ \& } C_3 \text{ are identical}).$$

11. We have AB = A

$\Rightarrow$ A (BA) = A [$\because BA = B$]

$\Rightarrow$ (AB) A = A

$\Rightarrow$ AA = A $\left[\because AB = A\right]$

$\Rightarrow$ A^2 = A

Again BA = B

$\Rightarrow$ B (AB) = B $\left[\because AB = A\right]$

$\Rightarrow$ (BA) B = B

$\Rightarrow$ BB = B $\Rightarrow$ B^2 = B

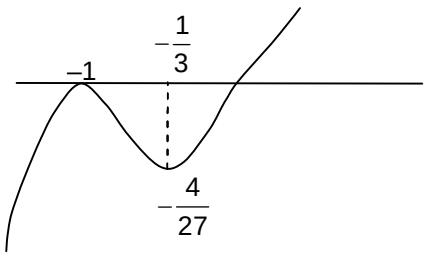
12. $|z| - \log_2(z + 2) = 0$

$z + 2 = 2^{|z|}$

$x + 2 = 2^x$ and y = 0

hence x = 2, y = 0.

13.



SECTION – C

1. $\tan^{-1} x + \tan^{-1} \frac{1}{y} = \sin^{-1} \frac{3}{\sqrt{10}}$

$\tan^{-1} \frac{1}{y} = \tan^{-1} 3 - \tan^{-1} x$

$\tan^{-1} \frac{1}{y} = \tan^{-1} \frac{3-x}{1+3x}$

$y = \frac{3x+1}{3-x}$ x=1,2 and corresponding y=2,7, thus two solution

3. $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$

$$4A = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$$

$$A^2 - 4A - 5I = 0$$

$$\begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0 \qquad A^2 - 4A - 5I = 0$$

Multiplying by A^{-1} both sides,

$$A^{-1}\left[A^2 - 4A - 5I\right] = A^{-1}0$$

$$A - 4I - 5A^{-1} = 0$$

$$A - 4I = 5A^{-1}$$

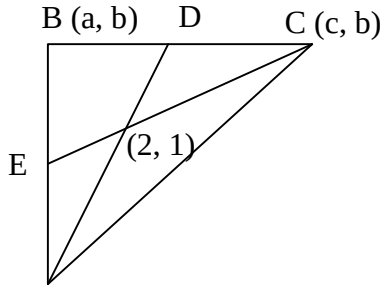
$$A^{-1} = \frac{1}{5}(A - 4I) = \frac{1}{5}\left(\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}\right)$$

$$A^{-1} = \frac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}.$$

4. Let equation of new plane $2x - 2y + z - 3 + \lambda z = 0$

Point (3, 1, 1) lie on $it \Rightarrow \lambda = -2$

Hence equation of new plane 2x – 2y –z = 3



$$\cos\alpha = \frac{4 + 4 - 1}{3.3} = \frac{7}{9}$$